

FIELD STUDY · NICE CONFERENCE & EXPO 2026

The Last Operator and the Philadelphia Convergence

*When a thesis developed in Brazil met, on the other side of the continent,
the confirmation of someone who lives the operation*

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Opening Note

In June 2026, I was designated by the Cyber Defense Commander to represent Brazil at the NICE Conference & Expo in Philadelphia — the foremost annual event on cybersecurity workforce development in the United States. I traveled with that mission and a range of professional expectations. But I carried one personal question in my luggage, and came back with something I had not anticipated: a confirmation.

For the past several months I have been developing a thesis I call *The Last Operator*. In one question, it asks: is artificial intelligence silently eliminating the professional who can still think from scratch? In cybersecurity, I argue this would not merely be a technical loss or a shortened learning curve. It would be a strategic vulnerability — because a nation that adopts automated systems without preserving professionals capable of understanding, auditing and questioning them loses, gradually and imperceptibly, cognitive sovereignty over its own security. Not because the machine fails, but because no one can verify whether it was right.

In one of the sessions of the pre-conference block — a closed workshop, small format, with little visibility beyond the room where it happened — I found much of that argument already formulated, independently, by Nik Alleyne: principal instructor at the SANS Institute, author of three books, and responsible for managed security operations for dozens of governments across the Americas. He did not know the EDR Triad. He did not know the concept of the phronimos. He did not know the CCOS. And yet, drawing on nearly three decades of practice, he arrived at conclusions remarkably close to those I have been building from Brazilian doctrine, philosophy and field experience.

This document was born from that encounter. It is not the account of an attendee who admired a speaker. It is the record of a convergence between two professionals who, departing from different worlds, described the same problem — and an attempt to show why that convergence matters, especially for Brazil. I first present what was said in the session, with the greatest possible fidelity. Then, where my thesis and Alleyne's remarks meet. Finally, what recent science has already demonstrated about all of this, and what it implies for our national strategy.

Three terms that will recur throughout the text

EDR Triad — governance framework I propose, built on three requirements: Explainability (algorithmic and operational), Algorithmic Due Diligence, and Accountability.

Phronimos — in Aristotelian ethics, the one who possesses phronesis: the practical wisdom to judge and decide well in concrete situations, distinct from mere technical competence.

CCOS — Cognitive and Situational Oversight Layer (Camada de Vigilância Cognitiva Situacional): the institutional function of keeping human judgment active and capable of questioning the automated systems it operates.

PART 1

The Event and the Session

The NICE Conference & Expo is the foremost annual gathering on cybersecurity workforce development in the United States. It is conducted under the NICE initiative — National Initiative for Cybersecurity Education —, linked to NIST within the U.S. Department of Commerce, and brings together leaders from education, government, industry, and non-profit organizations. The 2026 edition took place between June 1 and 3 at the Philadelphia Marriott Downtown, organized by Florida International University and New America, under the theme *From Foundations to the Future: Transforming the Cybersecurity Workforce*. The theme itself is already a diagnosis: the field acknowledges that its foundations are under pressure from the pace of technological change, and that the answer is not to abandon them, but to reintegrate them into the new reality.

The format and where the session fits

Understanding the event's format is necessary to correctly situate the session analyzed here. The first day, Monday June 1, is not the main conference: it is a block of pre-conference workshops, with separate registration, small groups, each workshop divided into two two-hour sessions with a networking break in between. It is a deliberately denser and more interactive space than the plenary sessions. The following two days, Tuesday and Wednesday, host keynotes, multi-speaker panels, and the Expo floor — the traditional auditorium and multi-track format. It was in the first block, the workshops, that Nik Alleyne presented.

The session was specifically part of the Conference Workshop Session 3, titled ***Securing the Cyber Workforce for Emerging Technologies: Workforce Trust Management and Readiness Across the Americas***. The title is revealing in two ways. The word *trust* is not decorative: it is the axis around which Alleyne organized his entire argument on AI accountability and governance. And the scope *Across the Americas* is not incidental: this edition hosted, in its pre-conference block, RICET — the Regional Initiative for Cybersecurity Education and Training —, an effort hosted by Florida International University and the Organization of American States in cooperation with NICE, aimed at building a common foundation of cybersecurity education and training across the region.

This hemispheric framing carries direct weight for Brazil, and helps explain why the session addressed managing the security posture of dozens of governments across the Americas. What was being discussed there was not an isolated American problem, but a continental challenge — one in which Brazil, by geography and institutional weight, is a natural stakeholder. Being in that room, representing the Cyber Defense Command, was also occupying the Brazilian seat in a conversation that was already happening about the future of our region.

PART 2

Who is Nik Alleyne

Nik Alleyne occupies an uncommon place in the field: he is neither the academic who never left the classroom, nor the practitioner who never stopped to reflect on what he does. He is both at once — and it is precisely that dual nature that makes his reading of AI and the workforce worthy of study. With over two decades in technology and the last seventeen years dedicated to security, his career rests on two pillars that rarely coexist with equal depth: real operations, as Head of Managed Security Services at an MSSP serving governments across the Americas, and elite technical instruction as a SANS Principal Instructor — the highest level in the SANS Institute instructor hierarchy, the global reference for security training.

At SANS, he teaches three courses that together cover exactly the terrain where AI is now advancing: SEC595 (Applied Data Science & AI/Machine Learning for Cybersecurity Professionals), which gives him the technical foundation to speak about AI without hype; SEC503 (Network Monitoring and Threat Detection In-Depth), which treats detection as a deep human skill; and SEC504 (Hacker Tools, Techniques, Exploits & Incident Handling), which anchors defensive judgment in offensive understanding. When someone who teaches applied machine learning, network detection and incident response warns about what is lost when AI occupies those three spaces simultaneously, it is worth listening: he knows in detail what automation is replacing. Add to that a Master's degree in Cyber Security Forensics and some of the most demanding certifications in the field — CISSP, GCIH, GCIA, GMLE and GPEN — as well as a well-regarded technical blog, securitynik.com.

The work: a journey from technique to critical awareness

Alleyne distributed three of his own books at the end of the session. Read in sequence, they describe a progression that, not by coincidence, mirrors this analysis's own argument. *Hack and Detect* (2019) establishes the base: using the Cyber Kill Chain model, it teaches that effective detection requires understanding the attack in depth, not merely recognizing signatures — an AI recognizes the pattern; the trained operator understands why the pattern exists. *Mastering TShark: Network Forensics* (2021) deepens traffic analysis using a command-line tool that offers no dashboard, no automatic alert, no recommendation: from it you only extract what you already know how to name. It is a deliberate celebration of manual skill as an irreplaceable foundation. And *A Little Book on Adversarial AI* (2025) marks the turn: it examines what happens when AI systems themselves become the target — data poisoning, model extraction, inversion attacks, prompt injection. It is the bridge between technique and thesis: adopting AI without understanding its attack vectors does not increase security; it merely shifts the attack surface to terrain the operator does not understand.

There is something significant in the fact that Alleyne himself embodies what he defends: an operator who built his judgment over decades, and who now speaks publicly about his concern for the generations arriving in the field without having walked that path.

PART 3

The Session, in Detail

The session was structured as an interactive tabletop exercise: twelve questions organized across three dimensions, answered in real time by the audience via QR code, with results projected live. Before the exercise, however, Alleyne built a dense conceptual foundation — and it is there that the richest material lies.

The wrong question and the right question

The session's strongest slide was not the most technical, but the most philosophical:

“Should we use AI in cybersecurity? That is the wrong question. The right question: how do we maintain accountability for the decisions AI makes on our behalf?”

Slide by Nik Alleyne, NICE Conference 2026

The question “should we use AI?” is already settled — AI already inhabits SIEMs, EDRs, threat intelligence platforms and SOARs. The real operational question is different: when AI decides, who answers? When it is wrong, who is capable of noticing and correcting? And who is training the professionals who will need to do that? Alleyne closed the argument with a triple requirement — that humans retain meaningful control, clear accountability, and the skills to govern what we have built. Without all three, automation does not increase security: it merely distributes vulnerability in a more opaque form.

Worth noting, without developing yet: that question — accountability for AI decisions — is exactly the third pillar of the EDR Triad, the Accountability pillar. I return to it in Part 4.

The EPOCH framework (MIT Sloan)

To ground the idea of human-machine complementarity, Alleyne drew on a 2025 MIT Sloan study by Isabella Loaiza and Roberto Rigobon: *The EPOCH of AI: Human-Machine Complementarities at Work*. The paper proposes five categories of human capabilities that AI cannot replicate, summarized in the acronym EPOCH:

	Capability	Description
E	Empathy	Emotional intelligence; reading human and relational context
P	Presence	Connection, networks and institutional presence
O	Opinion / Judgment	Forming opinions and judging based on ethics and experience
C	Creativity	Imagination, innovation, lateral thinking
H	Hope	Vision of the future, leadership, the capacity to project

The study’s central conclusion is direct: the future of productivity is augmentation, not automation. The authors explicitly refuse to call these “soft skills” — a technical skill, like solving a math

problem, is comparatively easy to teach; teaching judgment, empathy or vision is far harder. The data shows the market already rewarding those who have them: EPOCH-intensive occupations have grown more and project better. The market values judgment — while the training system produces it less and less. There is here the echo of a far older philosophical structure, that of the Aristotelian phronimos, which I return to in Part 4.

The three dimensions of the exercise

The tabletop organized the diagnosis across three axes. **Skills**: is the organization building the capabilities needed for an AI-enabled security workforce? The question is not whether tools exist, but whether people can operate them without losing autonomous judgment — having access to a tool is not the same as knowing how to evaluate, question or replace it. **Governance**: many organizations have adopted AI faster than they could develop governance frameworks — and that gap is precisely the space where risk accumulates undetected, where the dashboard stays green while no one can explain why. **Collaboration**: formal partnerships between jurisdictions, a topic that, in the Americas framing, speaks directly to Brazil.

AI for Security versus AI Security

One of the session’s most useful distinctions separates two fields that are often conflated: using AI to do security, and securing AI itself.

AI for Security — AI as a tool	AI Security — AI as a target
Phishing, intrusion and anomaly detection	Data poisoning and model poisoning
Threat hunting and log analysis	Inversion attacks (training data reconstruction)
Agentic defense solutions	Model extraction and model weight theft
Novelty detection	Prompt injection and AI infrastructure attacks

Most organizations that adopt AI for security do not consider that the model itself becomes an attack surface. Data poisoning is the perfect example: the corruption of training data does not appear on any conventional dashboard. The system runs, alerts fire, reports are generated — but the reference frame on which everything depends is compromised. It is the subject of Alleyne’s third book, and it converges with something I maintain in my thesis: AI does not “fail” in the trivial sense; it operates with precision on a reference frame that may be corrupted. It hits the wrong target with perfect accuracy.

This point has a dimension beyond the attack description: data poisoning can serve as a validation test for one of the thesis’s mechanisms. I return to this hypothesis in Part 4, when presenting the CCOS.

The judgment AI cannot transmit

The most revealing moment was not in the slides, but in the dialogue with the audience. When pressed about AI’s impact on the formation of junior professionals, Alleyne articulated — and here I paraphrase — an idea that captures the whole problem: someone with nearly thirty years in the field immediately recognizes when AI returns garbage, because they have been trained for decades to do exactly that; the professional who just left school does not. Not through personal

failure, but because we handed them language models and agentic AI without their having developed the criterion to evaluate what they receive. And the question that lingers is: who is teaching that judgment?

That evaluative criterion — the capacity to know when an output is inadequate without needing another tool to confirm it — is, in Aristotelian ethics, the core of phronesis. What Alleyne calls the ‘judgment factor’ is precisely what defines the phronimos. The convergence is addressed in detail in Part 4.

That “judgment factor” cannot be taught by slides. It forms over years of exposure to real errors, incidents that did not go as expected, situations where the tool said one thing and reality said another. AI does not merely execute faster: it removes from the learning process the friction that produced judgment. Alleyne closed this block with a set of rhetorical questions of surgical precision — where should automation stop and where does the organization’s legal and ethical responsibility begin; how to guarantee accountability (for him, you need *a named human attached to a decision threshold* — who takes the call at 2am when AI is wrong?); how to build auditability by design, not as an add-on; and why to distrust the premise that “AI will make us more secure,” which only holds if the governance layer preserves quality, accountability and human judgment at critical decision points.

The attentive reader will already have recognized, in that vocabulary — judgment, accountability, auditability, a named human answering for critical decisions — the contours of something I have been structuring independently. That is what the next part addresses.

The Convergence

What I witnessed in Philadelphia was not a thematic coincidence. It was independent confirmation, from someone who lives daily operations on another continent, of a diagnosis I have been building from Brazilian military doctrine, philosophy and field experience. Alleyne and I departed from distinct worlds and arrived at the same place — and it is the meeting of two independent paths that gives the conclusion its strength.

Same diagnosis, complementary scales

The Last Operator thesis holds that AI is silently eliminating the operator who thinks from scratch, and that this constitutes a national strategic vulnerability. Alleyne did not use those words, but said, in substance, the same: when he argues that the recently trained professional lacks the “judgment factor” because they formed operating AI-delivered outputs instead of building the reasoning that leads to them, he is describing the same extinction — seen from the angle of someone who teaches and operates, not from the angle of national sovereignty. The difference in scale is precisely what makes the two perspectives complementary: he looks inside the operator and the team; I look at the system and its consequences for the State.

	Alleyne's Diagnosis	The Last Operator Thesis
Scale	Individual and organizational	Strategic and national
Focus	The judgment deficit in the operator	The systemic vulnerability of that deficit
Stated cause	AI adoption without adequate training	Illusion of maturity produced by automation
Final risk	Absence of accountability	Loss of cognitive sovereignty
Response	Governance + ability to audit AI	EDR Triad and CCOS as public policy

The EDR Triad, said in other words

The EDR Triad is the governance framework I propose. Its four requirements appeared, one by one, in Alleyne’s remarks — without him naming them as such:

EDR Triad pillar	How it appeared in Alleyne's session
E ¹ — Algorithmic Explainability (interpreting the model's reasoning)	“Auditability must be a design decision, not a nice-to-have.”
E ² — Operational Explainability (the human understanding and questioning the output)	The “judgment factor”: the operator must know when the output is garbage.
D — Algorithmic Due Diligence (assessing risk before and during AI use)	“Challenge the assumption that AI will make us more secure.”

R — Accountability (who answers for the decision)	“We need a named human attached to a decision threshold.”
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That two independent paths — one theoretical-doctrinal, the other forged across three decades of operations — arrive at the same four requirements is, in itself, a strong indication that the diagnosis is correct. The clearest correspondence is that of pillar R: Accountability cannot be diffuse, cannot belong to “the organization” in the abstract, cannot belong to “the AI.” It must be anchored in an identifiable person, with the authority and competence to answer. That is precisely what Alleyne formulated with the image of the named human and the 2am phone call.

The CCOS, implicit in the session

The CCOS — Cognitive and Situational Oversight Layer — is grounded, in my thesis, in Endsley’s situational awareness model and proposes an explicit institutional function: keeping human judgment active over automated systems, not as bureaucratic supervision, but as a critical governance function. Alleyne did not name it, but described it when defending “augmented oversight”: AI recommends, the human reviews, and when there is divergence between the two, that divergence must be analyzed — the AI may be wrong, the human may be wrong, but the difference matters and must be understood. That function of analyzing divergence is the core of the CCOS: not the function of operating the system, but of preserving the capacity to question it. It is what separates the replaceable operator from the Last Operator.

This is where I return to the data poisoning mentioned in Part 3. In the context of the thesis, data poisoning ceases to be merely an attack vector to monitor and becomes a validation test for the CCOS itself. The reasoning is as follows: if a compromised model delivers a technically plausible but substantively wrong result, and the human cognitive oversight layer identifies it before it produces effect, then the CCOS has fulfilled exactly the function for which it exists. The AI’s failure, in that scenario, becomes paradoxically proof — empirical evidence that preserved human judgment is not redundancy, but safeguard. The decisive distinction is between a system that fails detectably and one that fails invisibly: the former is recoverable, the latter is catastrophic, and what separates them is precisely the presence of an operator capable of perceiving the divergence. Data poisoning thus offers a path to empirically test the thesis’s central hypothesis — something I intend to develop in dedicated work.

This is one of the most promising hypotheses for experimental validation of the thesis: a controlled environment in which deliberately poisoned models are submitted to operators with and without an active CCOS function would make it possible to measure, rather than merely argue, the value of preserved human judgment.

The phronimos and the EPOCH: the same intuition, twenty-three centuries apart

The philosophical core of my thesis comes from Aristotle. In the *Nicomachean Ethics*, the phronimos is the one who possesses not only theoretical knowledge (episteme) and the skill to apply it (techne), but phronesis — the practical wisdom to judge what to do in concrete situations, accounting for context and consequences. The MIT EPOCH framework is, in essence, an empirical and contemporary translation of the same intuition:

Phronesis	EPOCH (MIT, 2025)	In cybersecurity
Ethical judgment	O — Opinion / Judgment	Deciding before an ambiguous alert
Relational presence	P — Presence	Coordinating incident response across teams
Vision of the future	H — Hope / Leadership	Anticipating threats; building culture
Empathy with context	E — Empathy	Weighing the human impact of security decisions
Creation of solutions	C — Creativity	Responding to unprecedented incidents

What Aristotle called phronesis, MIT calls EPOCH; what Alleyne calls the “judgment factor,” I call the condition for the Last Operator’s existence. The languages differ; the problem is the same.

And there is the counterpoint. The Aristotelian deinotes is not an operator inferior to the phronimos in terms of knowledge or technical capability — Aristotle is explicit about this in Book VI of the *Nicomachean Ethics*. The deinotes possesses the same instrumental ability: the capacity to identify and execute the means required to achieve any given end. What distinguishes him from the phronimos is not what he knows. It is what he is. The deviation is one of character, not of competence. Without the ethos that orients that capacity toward good ends, technical skill converts into what Aristotle calls panourgia — mere cleverness in service of any purpose. In the context of contemporary cybersecurity, the deinotes is the professional who has the technical judgment, knows when the output is inadequate, understands the model’s limitations — but does not exercise that capacity because their character is not anchored in the collective good. They may be oriented toward efficiency, career advancement, compliance without friction, or delivering results that satisfy the manager even when they compromise real security. The deinotes’s risk is not that he errs. It is that he is precisely right in service of the wrong end.

The deinotes is the third figure that emerges from the thesis — distinct from both the phronimos and the operator degraded by automation. The degraded operator has lost judgment; the deinotes has it, but does not orient it by ethos. The three figures will be developed as a proper typology in subsequent work.

What Science Has Already Demonstrated

Until now, the convergence between Alleyne's session and The Last Operator thesis might be read as the coincidence of two experienced readings. But the phenomenon we both describe has, in recent years, moved from practitioner intuition to empirical research. It is worth anchoring the thesis in that literature — it is what separates it from opinion and moves it closer to a testable scientific hypothesis.

The erosion of critical thinking now has a measure

In 2025, researchers at Microsoft Research and Carnegie Mellon University published, at the CHI conference, a study with 319 knowledge workers and 936 real-world cases of generative AI use. The central finding: the more a person relies on AI tools, the less critical thinking they exercise — and the harder it becomes to engage that skill when it is needed. The study also found that users with AI access produced a less diverse set of outputs for the same task, which the authors explicitly interpret as deterioration of critical thinking. It is the first robust metric for what Alleyne describes in the field and what my thesis calls silent erosion.

The phenomenon has a name — and three decades of study in aviation

The mechanism behind this is not new. Human factors psychology has studied it since the 1990s under the names of automation bias and automation complacency — the operator's tendency to over-trust the automated system, ceasing to seek confirmatory or contradictory information. The literature was born in aviation, from accidents in which pilots trusted the autopilot to the point of failing to adequately monitor its functioning, with consequent loss of situational awareness — a concept I owe, in my thesis, to Mica Endsley, the same central author of that tradition. The decisive finding is that this body of research has now formally migrated to cybersecurity: a 2024 systematic review conducted at the University of Tulsa documents that increased automation in security operations centers (SOCs) amplifies exactly the same risk — analysts becoming dependent on automation and ceasing to exercise critical verification.

And it affects the youngest the most — those entering the field right now

There is still a finding that converges almost literally with Alleyne's warning about the recently trained professional. Recent studies on cognitive offloading — the delegation of cognitive tasks to external tools — find a negative correlation between frequent generative AI use and critical thinking capacity, and report that younger individuals exhibit greater dependence and lower scores. Precisely the operators entering the field right now, the same ones who, in Alleyne's words, "don't yet have the judgment factor," are the most exposed to erosion. The science confirms, simultaneously, his observation and my hypothesis.

Why this matters for the thesis

The Last Operator thesis ceases to be the intuition of a Brazilian officer and becomes a hypothesis aligned with converging evidence from Microsoft, Carnegie Mellon, NASA, the aviation human factors tradition, and systematic review on SOCs. I maintain it, even so, as a hypothesis under investigation — not as settled fact. But it is a hypothesis that recent literature supports, and does not contradict.

Strategic Implications for Brazil

The diagnosis presented in Philadelphia was delivered to a predominantly American audience. But the problem is universal — and in the Brazilian context, it takes on specific strategic dimensions that justify the Cyber Defense Command's representation in that room.

Brazil has the normative framework — implementation is what is missing

The country already has instruments pointing in the right direction. The National Cybersecurity Policy (PNCiber, Decree 11,856/2023) establishes, in Article 3, the objective of stimulating protection measures and risk management to prevent, mitigate and neutralize vulnerabilities — which directly implies operators capable of understanding, auditing and questioning the systems they operate, and not merely consuming automated outputs. The National Cybersecurity Strategy (E-Ciber, Decree 12,573/2025) and the Military Doctrine of Cyber Defense (MD31-M-07) constitute relevant institutional frameworks, and the Cyber Guardian Exercise — the largest in Latin America — represents a concrete initiative for workforce training and development in a realistic environment.

This normative process took a relevant step on the day this document is finalized. On June 30, 2026, the Cyber Defense Commander presented to the Federal Senate his considerations on PL 4,752/2025 — the Cybersecurity Legal Framework, which proposes the creation of the National Cybersecurity Agency (ANCiber). The debate around the regulatory framework brings Brazil closer to the questions of governance, accountability and workforce development that this analysis addresses — signaling that we are in a window of opportunity to anchor in law exactly the mechanisms that Alleyne and The Last Operator thesis describe as indispensable. I had the opportunity to advise the Commander in preparing for that hearing, which connects this document directly to the institutional moment we are living through.

Our gap is deeper than the American one

What Alleyne describes as an emerging problem in the US is, for us, structural. The United States has NICE as a standing national policy with annual revision cycles, the NIST AI RMF as a reference for AI governance, and the SANS/GIAC ecosystem as an elite training infrastructure. Brazil is building those pillars — PL 4,752/2025, with the proposal of the Cybersecurity Legal Framework and ANCiber, signals progress; the Cyber Guardian Exercise points toward the institutionalization of training. But the gap between the speed at which Brazilian organizations are adopting AI and the maturity of our governance and training mechanisms is even wider than that observed in the US. What for them is a warning, for us is an urgency.

In this context, ANCiber — if created under the terms of PL 4,752/2025 — emerges as a natural candidate to play, in Brazil, a role analogous to that which NIST plays in the United States: technical, normative and governance reference for the national cybersecurity ecosystem. And consequently, the agency could anchor an initiative equivalent to NICE — a structured workforce development policy, with competency frameworks, training programs and professional maturity metrics that today do not exist in a systematized form in the country. The debate now taking place in the Senate is not merely about regulation: it is about whether Brazil will have, in the coming

years, the institutional architecture necessary to prevent the Last Operator from disappearing through absence of policy.

It is here that the thesis takes on its national strategic dimension. This is not merely about training good professionals. It is about preserving the sovereign capacity to understand, audit, question and — when necessary — replace the AI systems operating in critical national infrastructure. Without the Last Operator, that capacity simply ceases to exist, and cyber sovereignty becomes dependent on black boxes that no one at home knows how to open.

It is worth noting that Brazil does not start from a weak position in terms of institutional commitment. In the International Telecommunication Union's Global Cybersecurity Index, the country ranks among nations with advanced maturity, particularly in the legal and organizational pillars — a reflection of the normative frameworks cited above. The thesis's point, however, is more subtle: institutional maturity measured by legal instruments and formal structures does not capture the silent erosion of operational judgment. A country may score well on normative commitment and, at the same time, be losing the human base capable of exercising that commitment in practice. It is precisely that gap — between formal maturity and real cognitive capacity — that The Last Operator thesis seeks to name and address.

PART 7

Synthesis: What I Bring Back from Philadelphia

I went to Philadelphia to represent Brazil at an event on the future of the cybersecurity workforce. I came back with a conviction that is now double: the central problem of the field in the years ahead will not be the sophistication of threats, nor even the pace of automation. It will be the preservation of human judgment — the capacity of an operator to look at a green dashboard and ask, with the competence to answer, whether it is green because everything is fine or because no one can read what lies underneath.

Nik Alleyne reached that problem through the door of operations and instruction, over nearly three decades. I reached it through the door of doctrine, philosophy and state-level cyber defense. That two such distinct trajectories converge on the same point — and that recent empirical research supports that point — gives me confidence that The Last Operator thesis does not describe a personal concern, but a real, measurable and addressable trend.

Addressable is the key word. The diagnosis, alone, is sterile. What I propose — the EDR Triad as a governance and training framework, and the CCOS as an institutional function — is an attempt at a response: giving name, structure and enforceability to what Alleyne described as a practical necessity. Where he says “we need a named human attached to a decision threshold,” the Triad responds with the Accountability pillar. Where he says “auditability must be design, not an add-on,” it responds with Explainability. The convergence is not only of diagnosis; it is also of direction.

I close with what I consider the synthesis sentence of the session, and which could open any document in my thesis:

“Your aim should not be to slow down automation, but to ensure humans retain meaningful control, clear accountability, and the skills to govern what we’ve built.”

Nik Alleyne, NICE Conference 2026

For Brazil, the question that follows is direct: are we training professionals with the skills to govern what has been built? Not to operate — to govern, audit, question and diverge when necessary. If the answer is “not yet,” then the EDR Triad and the CCOS are not academic exercises. They are instruments of public policy that we need to discuss now, before the gap becomes irreversible. That is the conversation I intend to keep carrying forward — and which, I hope, this record helps to provoke.

An acknowledgment

This document was also written as a gesture of recognition toward Nik Alleyne’s work, whose session in Philadelphia provided a valuable and unexpected confirmation of a line of research I am developing in Brazil. The convergence described here is, in my reading, a starting point for dialogue — between practices, between institutions, and between the Americas — that I intend to cultivate.

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